



FIRST SEMESTER 2026-2027
COURSE HANDOUT

Date: 01-08-2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No: CHEM F311

Course Title: Organic Chemistry-III

Instructor-in-Charge: ANIL KUMAR

Instructor: Anil Kumar

1. Course Description: Applications of important reagents and reactions in organic synthesis and disconnection or synthon approach will be emphasized in this course. Common organic reagents, Organometallic reagents, Transition metal catalyzed reactions, basic principles of disconnection, order of events, chemoselectivity, regioselectivity etc. introduction to retrosynthetic analysis using one group C–X and C–C disconnections, two group C–X and C–C disconnections, ring synthesis (saturated heterocycles), synthesis of heterocyclic compounds and complex molecules.

2. Scope and Objective of the Course:

- understand use of synthetic reagents.
- learn retrosynthesis of organic molecules.
- design the synthesis of organic compounds using appropriate reagents.

3. Course learning outcomes: At the end of the course, the student should be able to

- Apply common reagents in particular oxidizing and reducing reagents in organic synthesis
- Understand how organometallic reagents such as organoboranes, organosilanes, organotin, organoaluminums, Grignard and Gilman-type reagents are used to form C–C or C–heteroatom bonds and transform functional groups.
- Design the synthesis of organic compounds using appropriate reagents.

4. Text Books:

TB1: Michael B. Smith & Jerry March, Advanced Organic Chemistry, John Wiley & Sons, 6th ed., 2012.

TB2: Stuart Warren: Organic Synthesis: The Disconnection Approach: John Wiley & Sons, 2004.

5. Reference Books:

R1: Paul Wyatt, Stuart Warren: Organic Synthesis: Strategy and Control, Wiley; 1st edition, 2007.

R2: Morrison & Boyd, Organic Chemistry, 6th edition, Prentice-Hall, Inc., 1992.

6. Course Plan:

Module No.	Lecture/Tutorial Session	Reference	Learning outcome
1	L1: Introduction to organic synthesis and its applications L2-5: Oxidation: Metal based oxidizing reagents: Chromium, manganese, ruthenium, and silver-based reagents. Non-metal based oxidizing reagents: DMSO, peroxide and peracid, miscellaneous oxidizing reagents like IBX, DMP, CAN, DDQ, periodate, SeO ₂ ,	TB1: Selected topics & Class notes	Apply the use of common oxidizing and reducing reagents in converting one organic compound to another.



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	Tl(NO ₃) ₃ etc. L6-8 Reduction: Homogeneous/heterogeneous; hydrogenations, dissolving metal reductions.		
2	L9-L11: Organometallic reagents: Organoboranes, organosilicon reagents, organotin reagent and organoaluminum reagents	TB1: Chapter 2 & 4 Refer class notes	Apply the use of organometallic reagents in functional group conversions.
3	L12-L15: Transition metal catalyzed reactions: Carbon-carbon bond forming reactions (Suzuki, Heck, Sonogashira, Stille reaction), C–N bond forming reactions (Buchwald- Hartwig reaction)	TB1: Chapter 5 R1: Chapter 43	Devise synthesis of organic compounds using metal-catalyzed coupling reactions
4	L16-L18 Basic principles of disconnection approach in organic synthesis	TB2: Chapter 1 & class notes	Describe the general rules of disconnection approach & simple synthons
5	L19-L21: Synthesis of aromatic compounds: Basic principles, synthesis of aromatic compounds, the order of events.	TB2: Chapter 2, 3	Propose retrosynthesis of straight target connection of aromatic molecules
6	L22-L28: One group C-X and C-C disconnections, chemoselectivity, synthesis of alcohols, general strategy of choosing disconnection, stereoselectivity, synthesis of carbonyl compounds, regioselectivity, alkene synthesis, use of acetylenes for synthesis.	TB2: Chapters 4, 5, 10-16	Choose the realistic synthetic pathway after one group C-X/ C-C disconnection. Explain each stage of the proposed synthesis
7	L29-L35: Two group C-X disconnections, reversal of polarity, cyclisation reactions, summary of strategy, amine synthesis, Diels-Alder reactions, 1,3-difunctionalised compounds and α,β -unsaturated carbonyl compounds, control in carbonyl condensations, 1,5-difunctionalised compounds, Michael addition and Robinson annelation, use of aliphatic nitro compounds in synthesis, 1,2-difunctionalised compounds, FGA and its reverse, strategy of carbonyl disconnections.	TB2: Chapters 6-9, 17-28	Choose the realistic synthetic pathway after the two group C-X and C-C disconnection. Explain each stage of the proposed synthesis
8	L36-L38: Synthesis of three, four, five and six membered rings and general strategy of ring synthesis	TB2: Chapters 29, 30, 32, 34, 36, 37.	Devise synthesis of small-ring aliphatic compounds
9	L39-L40: Synthesis of aromatic heterocycles and advanced strategy	TB2: Chapters 39, 40	Devise synthesis of aromatic heterocycles



7. Evaluation Scheme

Component	Duration	Weightage	Date & Time	Nature of component (Close Book/ Open Book)
Quizzes*	20 min each	25%	Continuous	Close Book
Attendance based evaluation**	10 min each	10%	Continuous	Close Book
Mid-Semester Test	90 min.	25%	09/10 AN1	Close Book
Comprehensive Examination	180 min.	40%	14/12 FN	Close Book and Open Book

***THREE** quizzes (25 Marks each) will be conducted in entire semester. Best **TWO** will be considered for final evaluation.

****FOUR** tutorial tests will be conducted in the entire semester. The tutorial tests will be graded only if the student maintains a minimum of 50% attendance in lectures and tutorials.

7. Minimum criterion for awarding valid grades: Where a student continued to remain registered in a course but gave the instructor inadequate opportunity to evaluate him/her by absenting himself/herself from quizzes/ tests/ examinations/other components of evaluation, or by appearing in the same for the sake of appearance without applying himself/herself to the task in hand or by submitting a blank script (answer book) this event will be reported as NC (Not Cleared). **Moreover, a student should obtain 30% of the average of the top 10 performers in the class, or 40% of the median marks of the class, whichever is lower to clear the course.** If any student gets the marks lower than the prescribed standard mentioned above, he/she may be awarded NC.

8. Chamber Consultation Hour: Saturday, 12.00 PM-1.00 PM.

9. Notices: Notices concerning the course will be shared by email/displayed on Nalanda.

10. Make-up Policy: There will not be any makeup for any continuous evaluations. Make-up will be considered for Mid-Sem and Comprehensive Exams **only for genuine cases** (with valid documents of proof) where it will be felt that the student couldn't appear in the specific component of evaluation.

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